Is My Phone listening? A Mixed-Methods Investigation into the Belief in Conversation-Related Mobile App E-Eavesdropping in China

Shuai Cheng
Zhejiang University
cs36@zju.edu.cn

Haitao Xu(✉)
Zhejiang University
haitaoxu@zju.edu.cn

Junquan Wang
Zhejiang University
laotie666@zju.edu.cn

Xiaoqin Liang
Old Dominion University
xlian001@odu.edu

Shuai Hao
Old Dominion University
shao@odu.edu

Chuan Yue
Colorado School of Mines
chuanyue@mines.edu

Wenrui Ma
Zhejiang Gongshang University
mawenrui@zjgsu.edu.cn

Zhao Li
Hangzhou Yugu Technology
lzjoey@gmail.com

***Abstract*—Users widely suspect that mobile apps eavesdrop on face-to-face conversations for promotional purposes, presenting significant privacy concerns. Although technically feasible, such practices remain neither empirically verified nor falsified, creating a persistent tension between users' widespread reports and app vendors' consistent denials. However, little is understood about how these beliefs form and endure within underexplored non-Western contexts like China's, or how technical literacy influences belief formation under conditions of limited transparency. Hence we conducted a mixed-methods study involving Chinese users, combining in-depth interviews (N=21) with a survey (N = 203). Results show a high self-reported prevalence of perceived eavesdropping experiences (77.3%) and strong belief endorsement (74.4%), despite limited technical knowledge among participants. Beliefs were reinforced by greater technical awareness, distrust in vendors, and privacy cynicism, with technical explanations often strengthening suspicions. Stronger beliefs correlated with increased negative affect and reduced purchase intention. These results illuminate the emergence of surveillance beliefs and underscore the importance of enhanced user control over privacy mechanisms to restore trust.**

I. Introduction

The widespread integration of personalized advertising by mobile application (app) vendors has given rise to a deeply contentious issue: whether apps eavesdrop on users' offline conversations to gather private information for personalized advertising. This practice, termed *conversation-based electronic eavesdropping (e-eavesdropping)* [1], has neither been conclusively verified nor definitively refuted. Platform operators uniformly deny such activities [2–8], and security analysts argue that continuous audio surveillance would be economically and technically prohibitive compared to alternative targeting methods [8–10]. Nonetheless, users consistently report receiving advertisements closely matching their recent offline conversations [9, 11–13]. Media coverage of voice-to-text advertising patents [14, 15] further fuels public suspicion. Technical studies show that eavesdropping is feasible. For instance, non-acoustic sensors can capture speech-like signals [16–20] and privileged permissions could enable microphone access [21, 22]. However, feasibility does not prove actual deployment in mainstream apps, and empirical audits find no definitive evidence [9, 23]. Additionally, unlike input methods or clipboard monitoring, which is empirically documented in mainstream apps [24, 25] and typically involves user-initiated text input, conversation-based e-eavesdropping represents a distinct case where apps play no legitimate role in users' offline conversations. This practice remains *technically feasible but empirically unverified*, sustaining a fundamental tension between user testimony and vendor claims that constitutes the central dilemma in e-eavesdropping research.

Studies on e-eavesdropping and user perceptions show how ambiguity, mistrust, and everyday interpretation shape understandings of targeting technologies. Research on rumor and belief [26–28] shows that people use folk theories to explain opaque technologies, particularly when trust is low or experiences seem unusually personalized. Similarly, work on online behavioral advertising (OBA) [29–32] finds that opaque ad systems produce creepiness, suspicion, and misattribution. Studies of e-eavesdropping have primarily used surveys to examine self-reported experiences and predictors of belief [1, 11, 33, 34], with limited qualitative inquiry into how such beliefs form in everyday contexts [35]. Within broader dataveillance research, frameworks such as DEAL [36] offer comprehensive models for these perceptions by integrating informational cues, experiential triggers, and affective responses.

Several important limitations remain in the current literature. First, existing research on e-eavesdropping and surveillance beliefs is largely focused on Western [1, 34], leav-

✉ Corresponding author: Haitao Xu

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ing belief formation and evolution in non-Western settings underexplored, which motivates our focus on China as a theoretically relevant case. Second, prior studies primarily rely on survey [31, 33] rather than open-ended interviews, restricting insight into users' interpretive processes, contextual reasoning, and belief formation that structured questionnaires cannot fully capture. While qualitative work in related audio privacy research has explored experiential sense-making [35], Such approaches remain scarce in app-level e-eavesdropping research. Third, user concerns are often framed as misinformation or rumor, with limited attention to how *technical literacy* shapes reasoning. Despite expert observations that misattribution frequently arises when users conflate personalized targeting effects with actual e-eavesdropping [10, 13], existing studies rarely evaluate users' technical literacy regarding e-eavesdropping or personalization systems (e.g., personalized advertising or recommendation). This makes it difficult to determine whether e-eavesdropping beliefs stem from misunderstandings of targeting technologies or persist even when users correctly understand how such technologies function.

To address these limitations, we conducted a mixed-methods investigation in China, an underexplored non-Western context with one of the world's largest and most dynamic mobile internet user bases [37], where concerns about app eavesdropping are prevalent [8, 38, 39]. Rather than establishing systematic cross-cultural differences, we focus on how e-eavesdropping beliefs are formed and how they influence users within this setting. We explicitly incorporate users' technical literacy into our analysis, conceptualizing it along two complementary dimensions: knowledge of e-eavesdropping mechanisms and their limitations, and knowledge of alternative personalization pathways that can generate similar targeting effects without e-eavesdropping. The study is guided by four research questions:

- **RQ1:** What are users' perceptions and self-reported experiences of e-eavesdropping in China?
- **RQ2:** What is the level of Chinese users' objective literacy regarding personalized targeting in mobile apps?
- **RQ3:** What factors contribute to Chinese users' belief in e-eavesdropping?
- **RQ4:** What emotional and behavioral responses are associated with this belief?

Our investigation proceeded in three stages. First, a preliminary content analysis of user-generated posts and comments on Xiaohongshu examined how Chinese users perceive, interpret, and report app e-eavesdropping, providing descriptive evidence of user beliefs and self-reported experiences. Second, qualitative interviews with 21 participants explored personal experiences, perceptions, and the impact of technical explanations on belief formation. Findings revealed limited technical literacy, widespread distrust, and a paradoxical strengthening of beliefs following technical clarification. The interviews further indicated that knowledge of alternative personalization mechanisms played a more prominent role in belief change than knowledge of e-eavesdropping mechanisms and limitations. Third, a survey of 203 participants quantified these patterns, confirming high rates of self-reported experience (77.3%) and strong belief endorsement (74.4%), alongside limited understanding of indirect targeting mechanisms. Regression analysis identified six factors positively associated with belief strength, including our proposed construct of objective technical literacy, distrust, and privacy cynicism. Stronger beliefs correlated with increased perceived monitoring, negative affect, and reduced purchase intention. These qualitative and quantitative findings reveal a paradoxical role of objective technical literacy: greater knowledge of alternative personalization mechanisms may reinforce rather than correct e-eavesdropping beliefs under technical ambiguity.

In summary, our research makes the following contributions:

- **Comprehensive empirical study of e-eavesdropping perceptions, experiences, and responses in a non-Western context.** Using a mixed-methods design that integrates a preliminary technical assessment, semi-structured interviews (N = 21), and a large-scale survey (N = 203), we systematically document Chinese users' perceptions and self-reported experiences of suspected app-based e-eavesdropping. Our findings reveal high prevalence of such experiences and strong belief endorsement, quantify objective technical literacy, and demonstrate that stronger beliefs correlate with heightened perceived monitoring, negative emotional responses, and reduced purchase intention, highlighting practical implications for transparency, user control, and trust-building measures.
- **Identification of factors shaping e-eavesdropping beliefs in China.** Building on information- and experience-related factors identified in prior work, we propose and measure the construct of *objective technical literacy* and incorporate interview-derived factors such as *distrust* and *privacy cynicism*. Survey analyses further validate their significant associations with e-eavesdropping beliefs, revealing how knowledge, prior experiences, and attitudinal factors jointly shape e-eavesdropping beliefs and surveillance perceptions.
- **Revealing the paradoxical role of objective technical literacy in belief formation.** Moving beyond prior work relying on self-reported familiarity, we provide empirical evidence that objectively measured technical literacy can amplify e-eavesdropping beliefs rather than reduce them. Interview findings show that knowledge of alternative personalization mechanisms influences belief more strongly than knowledge of e-eavesdropping techniques themselves, often reinforcing existing suspicions. Survey results further reveal limited understanding of indirect data propagation in personalization mechanisms and a significant positive association between literacy and stronger beliefs. These findings challenge the assumption that literacy uniformly mitigates misconceptions, highlighting that increased technical awareness may paradoxically intensify surveillance beliefs under conditions of algorithmic opacity.

The remainder of this paper is structured as follows: §II reviews background on e-eavesdropping and related work. §III presents a preliminary analysis of Chinese users' self-reported experiences of app e-eavesdropping drawn from a major lifestyle-sharing platform. §IV and §V detail the methodology and findings of the qualitative interview study. §VI and §VII

describe the design and outcomes of the quantitative survey study. §VIII–§IX discuss the findings and conclude the paper.

II. Background and Related Work

This section reviews user studies on app e-eavesdropping and belief formation (§II-A), technical analyses and empirical measurements of potential e-eavesdropping (§II-B), and alternative personalization mechanisms with similar effects (§II-C).

A. User Beliefs and Interpretations of App E-Eavesdropping

Debates over app-based e-eavesdropping have been examined from technical and user-centered perspectives. Rather than assessing its technical existence, this subsection reviews empirical studies and theoretical frameworks on how such beliefs are formed, interpreted, and acted upon.

1) User Perceptions and Beliefs about E-Eavesdropping: Prior work has examined users' beliefs about e-eavesdropping primarily through survey-based approaches. Segijn et al. [1] conducted a cross-national survey (N = 886) across the United States, Netherlands, and Poland, showing that many users report ads linked to offline conversations and believe devices are listening, with such beliefs associated with negative emotions; however, the study is largely descriptive and does not examine the factors associated with these beliefs. Subsequent work by Segijn et al. [34] extends this line by identifying folk-theoretic explanations (e.g., e-eavesdropping, priming, coincidence, digital trace inference) shaped by contextual cues such as temporal proximity and cultural–regulatory factors, yet but does not objectively examine how technical knowledge shapes belief formation. Frick et al. [33] identify predictors of perceived surveillance (e.g., trust in devices, computer anxiety, prior experiences), but rely solely on quantitative methods and do not examine technical dimensions of belief formation. Complementing this, Dunbar et al. [35] qualitatively examine audio-related privacy perceptions and behaviors, characterizing users as Guardians, Pragmatists, and Cynics and deriving privacy-supporting design recommendations. However, their focus is audio privacy rather than the formation of beliefs about unverified app-level e-eavesdropping.

2) Theoretical Perspectives on Surveillance Belief Formation: Prior work has examined surveillance belief formation in broader contexts of privacy, data collection, and algorithmic systems. Early privacy research often assumes that increased technical literacy and transparency improve user understanding and control. Subsequent work by Acquisti et al. [40] and Brandimarte et al. [41] challenges this view, showing that privacy judgments are shaped more by heuristics, affect, and perceived control than by objective knowledge. Research on folk theories further suggests that users interpret opaque systems through partial knowledge and analogies: Diakopoulos et al. [42] argue that transparency may enhance perceived accountability without improving intelligibility, while Siles et al. [43] and DeVito et al. [44] show that users construct plausible explanations and infer intentionality from perceived system capabilities under low verifiability. These cognitive patterns are reinforced by broader social dynamics. DiFonzo et al. [45] show that partial knowledge increases certainty under ambiguity and distrust, while Draper et al. [46] found that users oppose surveillance yet accept it due to perceived inevitability. From a normative perspective, Nissenbaum et al. [47] emphasize that concerns arise from violations of contextual appropriateness, whereas Aguirre et al. [48] argue that perceived benefits and trust can legitimize data collection, although platform opacity may undermine such trust.

3) Theoretical Framework: Strycharz et al. [36] extend traditional advertising theories to digital contexts through the DEAL (Dataveillance Effects in Advertising Landscape) framework, which explains how users perceive, interpret, and respond to dataveillance, with surveillance belief as the core construct. The DEAL model comprises following components:

- **Surveillance Beliefs**: Foundational perceptions of surveillance scope and purpose [49], formed via (1) *information-driven* processes (e.g., transparency disclosures, privacy education, media) and (2) *experience-driven* processes (e.g., targeted-ad encounters, secondhand accounts, folk theories).
- **Perceived Surveillance**: The subjective sense of being monitored, triggered by *surveillance episodes* (e.g., ads revealing personal data use) that reinforce surveillance beliefs.
- **User Responses**: Responses driven by surveillance beliefs and perceived surveillance, including (1) *surveillance responses* (cognitive, emotional, and behavioral reactions such as reduced disclosure or privacy protection) and (2) *advertising responses* (e.g., purchase intention, perceived threat, reactance), with the former influencing the latter.
- **Boundary Conditions**: Moderating factors affecting response strength, including individual differences (e.g., privacy calculus, conspiracy mentality) and contextual aspects of dataveillance (e.g., timing, data type, purpose).

Our study adopts the DEAL model by conceptualizing app e-eavesdropping beliefs as surveillance beliefs, and examines their determinants and effects on user responses.

B. Techniques and Mechanisms of App E-Eavesdropping

Eavesdropping techniques in mobile apps fall into acoustic and non-acoustic sensors (e.g., optical, motion) [50, 51].

1) Acoustic Sensor-Based E-Eavesdropping: Acoustic e-eavesdropping primarily relies on microphones to capture audio. While many apps legitimately request microphone access like voice input, 43.8% of 17,000 popular Android apps were found to request audio-recording permissions [23]. Modern mobile OSes have introduced restrictions to limit covert recording. For example, Android 9 blocks background microphone access [52], while Android 12 and iOS 14 provide visual indicators to limit covert recording [53, 54]. However, privileged apps (e.g., system apps or voice assistants) can bypass these controls via APIs like `VoiceInteractionService` and `AlwaysOnHotwordDetector` [55, 56], and may not always trigger privacy indicators [57, 58]. Furthermore, third-party apps can exploit system vulnerabilities to obtain equivalent privileges, enabling covert eavesdropping via injection attacks or zero-day exploits. Earlier research demonstrated

unauthorized voice eavesdropping through mobile system vulnerabilities [21], and the 2023 `Pinduoduo` case illustrated how app exploits could bypass Android protections [22]. Such threats remain relevant: the 2025 LandFall campaign exploited a Samsung Android zero-day (CVE-2025-21042) with microphone and call-monitoring capabilities [59], alongside other recent Android zero-days, including CVE-2025-21043 in Samsung devices [60] and CVE-2024-53104 in Android [61]. Collectively, these cases demonstrate that while platform-level defenses restrict acoustic eavesdropping, privileged or exploited access preserves its technical feasibility.

2) Non-Acoustic Sensor-Based E-Eavesdropping: Recent work explores microphone-free eavesdropping via side channels, leveraging optical, motion, and radio-frequency (RF) sensors to infer speech without microphone permissions. Optical approaches reconstruct audio from vibration-induced visual signals (e.g., *Visual Microphone*, *lamphone*) but require line-of-sight, controlled lighting, and complex setups [17, 62, 63]. Motion-based methods use accelerometers and gyroscopes, which require no permissions, to capture speech from device vibrations (e.g., *Gyrophone*, *AccelEve*), offering stealth but limited accuracy due to low sampling rates and noise [16, 18, 64, 65]. RF-based techniques analyze sound-induced perturbations in wireless signals (e.g., *WiHear*, *mmSpy*), enabling non-line-of-sight sensing [19, 66, 67]. Despite expanding the attack surface, these approaches require specialized hardware, complex processing, or constrained environments, and there is no evidence of deployment in mainstream mobile apps.

3) Technical Feasibility Analysis: An analysis of 17,000+ Android apps found no systematic evidence of covert audio recording, with microphone use aligned with functionality [23]. A subsequent review proposed a threat model involving OS providers, apps, and libraries, identifying acoustic and non-acoustic eavesdropping as key vectors [9]. While technically feasible, there is no evidence of large-scale deployment.

C. Alternative Mechanisms to App E-Eavesdropping in Personalized Targeting

1) Technological Foundations: Personalized targeting systems collect and integrate multidimensional user data to build profiles for precise content delivery, which may be perceived as functionally similar to e-eavesdropping due to comparable privacy implications. The main approaches include: (1) **Behavior-Driven Targeting**, which uses user activity (e.g., browsing, clicks, search, purchases) to infer interests and personalize content [68]; (2) **Cross-Device Tracking**, which links user activity across devices via shared IPs, cookie synchronization, and logins to construct unified profiles [69–72]; (3) **Device Fingerprinting and Contextual Targeting**, which derives unique identifiers from device attributes and leverages contextual signals (e.g., location, Wi-Fi, Bluetooth) for persistent, cookie-free tracking [73, 74]; and (4) **Input Method-Based Data Collection**, where IMEs may collect and transmit keystroke data, enabling sensitive information leakage and expanding targeting data sources [24, 25, 75, 76].

2) User Perceptions and Interactions with OBA: Usable privacy research shows that users struggle to understand and control OBA. Ur et al. [29] found that targeted ads are often perceived as "creepy," especially when tied to sensitive or recent topics. Leon et al. [30] showed that opt-out tools (e.g., AdChoices) are difficult to use, limiting control. Chanchary et al. [31] further revealed fragmented mental models and mismatches between user understanding and actual practices. Recent work by Jane et al. [32] indicates that even improved transparency tools suffer from poor usability, fostering misunderstanding and cynicism. These findings show that users often have incomplete knowledge of OBA and limited control over it, which may also contribute to perceived e-eavesdropping.

III. Motivation

A primary motivation for this study was the observation of frequent user-reported perceptions and experiences of suspected app-related e-eavesdropping on Xiaohongshu [77], China's leading lifestyle-sharing and social-networking platform, where users discuss a wide range of topics including shopping and consumption. To systematically examine these self-reported accounts, we collected 10,250 related posts (2023–2025) via keyword matching (e.g., "eavesdropping," "app monitoring," "recommendation after talking"). After filtering and deduplication by user ID, we retained 1,921 posts from unique users. The posts were collected using the ScrapeGraphAI crawler [78], which retrieved publicly accessible Xiaohongshu content matching the predefined keyword set.

We adopted LLM-assisted topic analysis over traditional NLP methods, as Xiaohongshu posts are often colloquial, noisy, and context-dependent, requiring semantic interpretation beyond lexical co-occurrence. We selected GPT-5.4 for its contextual understanding and instruction-following capabilities, and designed prompts with explicit task descriptions and output constraints to improve classification consistency for informal content. A topic taxonomy was generated through iterative prompt-based extraction: We independently applied the topic-extraction prompt four times to generate candidate taxonomies, which were manually reviewed and consolidated by merging semantically overlapping categories into the final taxonomy. Each post was classified into one predefined topic using the final taxonomy. The extraction and classification prompts are provided in Figures 2 and 3 in Appendix B.

To validate our approach, we compared three prompt strategies on a manually annotated subset of 300 posts: a basic prompt containing only the task and topic names, a definition-guided prompt with topic descriptions, and our final structured prompt with explicit output constraints and classification instructions. For each strategy, we calculated Cohen's kappa between model-generated classifications and manual annotations, obtaining 0.61, 0.70, and 0.81, respectively. The results show that richer task definitions and output constraints improve classification consistency. This validation also reduces potential LLM-induced biases, such as sensitivity to prompt variations and preference for salient or frequent themes, although such biases cannot be fully eliminated. Therefore, we interpret the

taxonomy as a structured characterization of the collected posts rather than an exhaustive or unbiased representation of all e-eavesdropping-related discourse on Xiaohongshu.

Table VII in Appendix C outlines distinct categories of concerns about app-related eavesdropping, revealing specific firsthand experiences that trouble users. *Conversation Eavesdropping (*i.e.*, e-eavesdropping)* is the dominant fear, cited in over a quarter (28.3%) of instances, where users are convinced apps constantly listen through the microphone. This feeds into the second most common sentiment: *Pervasive Fear and Helplessness Towards Surveillance* (12.3%), where users express resignation and a sense of lost privacy. A particularly alarming belief for many users is *Thought Intrusion or Brainwave Monitoring* (11.9%), the idea that devices can access unspoken thoughts, as when a product considered mentally appears in a shopping app. These core anxieties are often fueled by users' perceived evidence of *Permission Abuse and Background Surveillance* (8.9%), where users find apps accessing microphones or cameras without consent. Finally, the seamless transfer of information prompts concerns over *Cross-Application Data Sharing and Tracking* (8.5%), where an activity in one app triggers a related recommendation in another, creating a perception of a unified, inescapable monitoring ecosystem. Such findings informed the design of our subsequent user studies.

IV. Qualitative Interview Methodology

We conducted interviews with 21 participants to investigate how users form and interpret e-eavesdropping beliefs, how technical explanations influence these beliefs, and what factors shape their perceptions.

A. Participant Recruitment

Participants were recruited through campus flyers and a university-affiliated research recruitment website; specific URLs have been omitted to preserve anonymity. Eligibility criteria required age over 18, smartphone experience, and exposure to in-app advertising. Age and gender quotas ensured demographic diversity. Each interview lasted 30-40 minutes, with participants receiving approximately 20 USD compensation in accordance with local ethical guidelines in China. The study received approval from our university's Institutional Review Board. Detailed demographic information of all participants (labeled P1–P21) are provided in Table VIII in Appendix D. The sample was distributed across age groups, closely reflecting the demographic composition of mobile users in China [37]. Gender distribution was balanced, with 11 male and 10 female participants. As participants were recruited from the university context, the sample is relatively well-educated, which may introduce bias in interpreting e-eavesdropping experiences. Thus, the findings provide explanatory insights into belief formation rather than population-level estimates.

B. Interview Structure

The interview consisted of three phases. In Phases I and III, we investigated participants' experiences, belief, and perceptions of e-eavesdropping. In Phase II, we presented the current technical landscape related to this phenomenon. The detailed interview procedure was structured as follows:

- **Phase I** examined participants' existing e-eavesdropping beliefs. At the beginning of each interview, we provided a definition of e-eavesdropping consistent with §I. Participants then were asked to describe any firsthand or secondhand experiences related to suspected e-eavesdropping and to elaborate on their beliefs regarding whether apps engage in such practices, including their reasoning.
- **Phase II** introduced participants to the technical background and current developments in e-eavesdropping. To examine both components of objective technical literacy, the briefing covered two complementary forms of knowledge: *e-eavesdropping knowledge* and *alternative-personalization knowledge*. The former included: (1) the classification, feasibility, and limitations of e-eavesdropping technologies in §II-B; (2) existing detection efforts in §II-B3; (3) official denials issued by major app vendors in China [3, 4, 6–8] and security experts [8–10]; The latter covered (4) other existing user information collection mechanisms supporting personalized advertising in §II-C. For non-acoustic attacks, we clarified that they remain research-confined, with no evidence of deployment in mainstream mobile apps.
- **Phase III** reassessed participants' e-eavesdropping beliefs and inquired whether and why their views had changed following the technical introduction. Participants were also asked about their prior unawareness with the content presented in Phase II. Finally, participants shared perceptions and responses toward e-eavesdropping or alternative personalized advertising practices, along with suggestions for app vendors regarding privacy protection.

The Phase II briefing was designed to provide a balanced overview of e-eavesdropping technologies rather than imply that technically feasible attacks are deployed in practice. It aimed to improve participants' understanding of the technical landscape and examine how technical literacy influences belief formation. We acknowledge that introducing unfamiliar attack scenarios may increase their salience and perceived plausibility. To mitigate this risk, the briefing distinguished technical feasibility from real-world deployment and balanced attack descriptions with their limitations, detection efforts, vendor denials, and alternative explanations. We therefore interpret subsequent belief changes as responses to the technical briefing, while recognizing potential briefing-induced bias.

To minimize participant fatigue and cognitive load, the briefing was limited to five minutes and delivered verbally using accessible language and concrete examples. Participants could request clarification, and the interviewer adjusted the pace when needed. Across the 30–40-minute interviews, we observed no apparent signs of cognitive overload, such as difficulty following the briefing or subsequent discussions. The complete interview protocol is provided in Appendix E.

C. Data Analysis

We analyzed the interview data using thematic analysis. All interviews were manually transcribed and coded using NVivo

[79]. Two researchers independently coded the transcripts and developed initial codebooks, which were subsequently merged through discussion and adjudication by a third researcher. Following five rounds of independent coding and iterative codebook refinement, inter-coder reliability was assessed on a random subset of transcripts independently coded by both researchers using the final codebook (Appendix F), yielding a Cohen's κ of 0.91 [80]. To minimize meaning loss from translation, all coding and thematic analysis were conducted in Chinese. Only the interview excerpts in §V and theme labels in Appendix F were translated into English. Both coders were proficient in Chinese and English, and all translated materials were cross-checked for accuracy and consistency.

V. Qualitative Interview Findings

We present findings following the interview structure. §V-A covers e-eavesdropping experiences and initial beliefs (RQ1); §V-B examines technical understanding and belief evolution (RQ2); §V-C–§V-D discuss user perceptions, privacy concerns, emotional and behavioral responses (RQ3–RQ4); and §V-E presents user suggestions. Interview questions (IQs) are listed in Appendix E.

A. Experiences and Belief of E-Eavesdropping (RQ1)

Among the 21 participants interviewed, 11 reported firsthand experiences of suspected app e-eavesdropping (IQ6), while 6 described secondhand encounters (IQ7). Regarding beliefs (IQ8), 15 participants reported believing that app e-eavesdropping exists, while 6 reported not. Among the six non-believers, five attributed their skepticism to a *lack of personal experience*, and one considered e-eavesdropping *technically infeasible*. Participants who believed in e-eavesdropping provided several rationales:

- **Reliance on Firsthand Experiences** (10 participants in total): Many participants inferred the existence of e-eavesdropping based on personal encounters. For instance, P18 remarked: *"I believe e-eavesdropping exists because before we discussed a product, it rarely appeared in apps. Yet shortly after mentioning it, I frequently see exactly those products when browsing an app."*
- **Trust in Secondhand Experiences** (P4): A participant adopted beliefs based on others' reported experiences. P4 noted: *"Many people around me have observed this phenomenon; it cannot be mere coincidence."*
- **Analogy to Existing Technologies** (P10, P11): Several participants, familiar with other forms of personalized advertising systems, assumed e-eavesdropping would be technically feasible. P11 commented: *"Since apps can monitor text input, they should also be capable of monitoring my voice."*
- **Distrust of App Vendors** (P5, P7): Two participants expressed general skepticism toward developers. P5 argued: *"It is easy for developers to incorporate backdoors into programs, and I doubt many strictly adhere to regulations."*

Interpretation: These responses indicate that participants actively constructed causal explanations to resolve ambiguity in personalized advertising, rather than merely recounting isolated incidents. Firsthand experiences provided salient cues, while secondhand accounts socially reinforced beliefs [28]. In the absence of transparent explanations, temporal proximity between conversations and subsequent ads was interpreted as evidence of intentional listening, reflecting attribution under technical ambiguity [81]. This pattern of reasoning, grounded in experiential cues, aligns with prior findings [1, 30]. Furthermore, analogies to existing technologies and distrust of app vendors suggest that e-eavesdropping beliefs emerge as part of broader sense-making under algorithmic opacity, consistent with research on folk theories [43] and digital resignation [46].

Finding I-1: Most participants believed in the existence of e-eavesdropping, with half reporting firsthand experiences, emerged as the primary driver of these beliefs.

B. Technical Literacy and Its Influence on E-Eavesdropping Beliefs (RQ2)

1) Lack of Technical Awareness (IQ9): After introducing the technical aspects of e-eavesdropping and personalized advertising, participants identified prior knowledge gaps; all except P3 reported some level of unawareness. These gaps fall into three categories: (1) *Complete unawareness* (8 participants), where users lacked understanding of underlying mechanisms and relied on experiential inferences, as P13 noted *"The explanations you provided are entirely new to me"*; (2) *Partial unawareness of personalized advertising technologies* (8 participants), where users knew basic mechanisms (e.g., click tracking) but were unaware of others such as input methods or network-based profiling, e.g., P5 expressed surprise at input method data collection; (3) *Unawareness of listening technologie*s (10 participants), where users were unfamiliar with microphone permissions or sensor-based e-eavesdropping, as P9 remarked: *"I had not considered that gyroscopes could potentially facilitate e-eavesdropping."*

2) Belief Evolution and Influencing Factors Following Technical Explanations (IQ10): Following the technical overview, 14 participants revised their e-eavesdropping beliefs, including 9 who *increased* their belief and 5 who *decreased* it relative to their pre-briefing views. Among those with increased belief, three previously skeptical participants (P2, P14, P21) shifted toward belief. The remaining 7 participants maintained their original views, with P8 finding the explanations reinforced the belief that e-eavesdropping does not exist, while P20 continued to rely on firsthand experience despite the lack of evidence.

Participants attributed these belief changes to two main factors, corresponding to the two components of objective technical literacy. First, 10 participants cited the *introduction of alternative personalization technologies*: 7 reported increased e-eavesdropping beliefs, as P4 noted: *"The mechanism of targeted advertising is just as malicious as e-eavesdropping. If apps can collect data in this way, they can certainly eavesdrop."* And P5 stated: *"Now I suspect even input methods might be e-eavesdropping."* In contrast, 3 participants (P1, P17, P11) treated personalization as an alternative explanation that decreased such beliefs, as P1 attributed the effect to data

synchronization across contacts. Second, 4 participants emphasized *technical feasibility and limitations of e-eavesdropping*: 2 (P6, P14) reported increased beliefs, with P14 stating: *"Now I think they do have software listening to us; it just hasn't been detected yet."* In contrast, 2 (P10, P15) reported decreased beliefs due to perceived technical constraints, as P10 remarked: *"From the e-eavesdropping techniques you introduced, I feel the deployment costs are just too high."*

3) Interpretation: These findings suggest two distinct pathways through which technical literacy influences belief. Knowledge of e-eavesdropping technologies shapes judgments of technical plausibility: awareness of feasibility may reinforce belief, while recognition of cost and implementation constraints may weaken it. In contrast, knowledge of alternative personalization mechanisms often reinforces existing suspicions, even when it could serve as alternative explanations.

Technical information is assimilated into pre-existing beliefs rather than processed neutrally, consistent with confirmation bias. Drawing on our qualitative data, we identify three complementary mechanisms: (i) familiarity without backend understanding; (ii) confirmation bias, where technical knowledge reinforces existing suspicions; and (iii) inference from feasibility to reality, where users assume that possible capabilities are actively deployed. This reflects opacity-driven sense-making, where partial knowledge increases certainty, as documented in work on folk theories and rumor formation [43–45]. It also contrasts with the information-deficit model, which assumes transparency improves understanding [42, 82].

A key implication is that knowing what systems can do (*capability literacy*) does not necessarily help users infer which mechanism caused an observed outcome (*causal-attribution literacy*). Current literacy efforts may therefore fail less from insufficient coverage than from not distinguishing technical feasibility, operational deployment, inferential targeting, and empirical evidence. Interventions should clarify these distinctions rather than simply exposing users to more mechanisms. Nevertheless, these interpretations should be qualified by potential briefing-induced bias, which may have heightened the salience and plausibility of the technologies discussed.

Finding I-2: Most participants lacked relevant technical knowledge. Technical explanations reshaped their beliefs: personalization knowledge more often strengthened beliefs, whereas eavesdropping knowledge had mixed effects.

C. User Perceptions and Evaluative Dimensions (RQ3, RQ4)

When examining participants' perceptions of app e-eavesdropping and personalized advertising they attributed to it (IQ11), we identified key concepts reflecting both belief formation (RQ3) and user responses (RQ4), ranging from negative to positive and ordered by frequency:

- **Distrust** (11 participants): Many distrusted advertising systems and vendors. P17 raised concerns about loyalty penalties: *"Prices are higher for frequent buyers."* P3 worried about data misuse: *"My private information could be sold to criminals."* Others questioned corporate claims, as P9 noted companies tend to protect their image, and P19 trusted personal experience over official statements.
- **Privacy Cynicism** (7 participants): Participants expressed resignation toward privacy loss. e.g., P7 stated there is no solution to user data leakage, while P9 accepted inevitability: *Since I cannot stop using these apps or prevent them from collecting data, I have no choice but to accept it."*
- **Negative Emotions** (6 participants): Responses included annoyance, fear, and worry. P2 found ads annoying, and P10 felt alarmed: *Apps know too much about me."*
- **Perceived Low Quality** (4 participants): Some criticized ad relevance and timeliness. P21 noted delayed recommendations, while P6 found ads misaligned with their needs.
- **Intrusiveness** (3 participants): Participants felt ads invaded personal space. As P17 stated: *'The app took my private information without consent and pushed content that made me aware of it, which I found offensive."*
- **Feeling Dominated** (2 participants): Some felt controlled by apps. As P10 explained: *'The app monitors me and tries to confine me in an information cocoon."*
- **Positive Evaluations** (9 participants): Others emphasized convenience and relevance. P13 valued efficiency in reducing browsing effort, while P12 appreciated precision: *I feel the advertising are getting more precise. Each time, they include items I genuinely want to buy, and that excites me."*

Interpretation: Participants' perceptions across these dimensions were closely interrelated, consistent with §III. Distrust, privacy cynicism, and negative emotions reinforced each other, making ambiguous targeting readily attributable to covert surveillance. Distrust made e-eavesdropping intuitive under uncertainty, while cynicism reflected resignation to data misuse, further reinforcing surveillance interpretations [83–85]. This reflects broader sense-making in opaque systems, where users infer intentionality behind algorithms [29], consistent with prior studies [43, 46]. Frustration, resignation, and perceived loss of control heightened surveillance concerns and reduced receptivity to advertising attributed to e-eavesdropping. These findings align with DEAL, where negative affect and reduced control correspond to surveillance responses, while diminished ad receptivity represents a downstream advertising effect [36]. Positive evaluations were insufficient to offset negative appraisals, indicating e-eavesdropping beliefs are embedded in a distrustful interpretive frame.

Finding I-3: Participants' evaluations of e-eavesdropping and related personalized advertising were predominantly negative. Among these, *distrust* and *privacy cynicism* emerged as the two most prominent evaluative dimensions.

D. User Responses to E-Eavesdropping and Personalized Advertising (RQ4)

We next examined participants' behavioral, emotional, and privacy-related responses to suspected e-eavesdropping, including their engagement with personalized advertising they attributed to such surveillance.

1) Clicking and Purchase Intentions (IQ12): When asked about responses to personalized advertising, seven participants refused to click, citing lack of need or resistance, as P16 preferred intentional search and P9 rejected perceived benefits: *"Such personalized advertising offer no genuine convenience."* Three participants clicked without purchasing, often for comparison, as P17 checked products but compared prices elsewhere due to loyalty concerns. Others showed random behavior, with P8 noting that clicking depended on mood.

2) Changes in App Usage Practices (IQ13): Only six participants reported modifying app usage due to privacy concerns, adopting strategies such as using new numbers, disabling microphones, reducing usage, limiting shared information, and restricting permissions. P12 described partial disclosure: *"If apps require multiple real details, I will provide one false entry."* Others did not change behavior, citing privacy cynicism or minimal perceived impact, as P20 remarked: *"Even with some privacy loss, I feel it doesn't impact my life much."* Functional constraints also limited change, as P4 noted: *"Without certain permissions, apps simply won't work."*

3) Concerns about Privacy Risks (IQ14): Participants identified several feared outcomes from user data leakage:

- **Personal Safety** (5 in total): Concerns included exposure of sensitive information such as home addresses leading to physical threats. P7 explained: *"If others knew my address, they could come to my home and harm me."*
- **Illegal Activities** (5 in total): Participants feared identity theft or misuse of personal data. P12 stated: *"Others might impersonate me to apply for loans."*
- **Financial Loss** (5 in total): Risks included account theft and fraud, as P10 worried: *"If my bank password is captured by the input method, my account could be stolen."*
- **Spam Calls** (5 in total): Many reported the nuisance of frequent telemarketing calls, as P14 said: *"Other risks are rare, but constant call harassment is already a reality."*
- **Misguided Consumption** (3 in total): Some feared unnecessary purchases driven by advertising. P17 noted: *"Repeated advertising of unnecessary item leads to wasteful spending."*
- **Confidentiality Breach** (2 in total): A few worried about work-related data leaks. P15 explained: *"Confidential documents on my phone could be accessed by apps, leading to a breach of confidentiality."*

4) Interpretation: These behavioral and emotional patterns indicate that perceived surveillance risk did not reliably motivate protective action. Hesitation to click, heightened vigilance, and fear suggest that e-eavesdropping beliefs primarily shape responses through affective and behavioral pathways. These patterns align with the surveillance-response dimension of the DEAL framework: fear and vigilance reflect affective responses, while avoidance and privacy-protective actions correspond to behavioral responses [36]. Yet the limited adoption of protective measures suggests that users feel monitored but view resistance as futile, owing to technical barriers and platform dependency—consistent with prior work on privacy cynicism and digital resignation [46, 85]. This combination of concern and inaction helps explain both the persistence of e-eavesdropping beliefs and their impact on advertising effectiveness, even in the absence of verified surveillance. At the same time, this pattern contrasts with privacy-calculus and empowerment accounts, which assume that risk awareness motivates protective behavior [40, 82].

Finding I-4: Participants' responses showed a gap between negative attitudes and protective actions: while many avoided engaging with personalized advertising attributed to e-eavesdropping, few adopted privacy protections, often reflecting privacy cynicism. Key concerns centered on personal safety, illegal use, financial loss, and spam calls.

E. User-Proposed Mitigation Strategies

Finally, participants were invited to provide suggestions for app vendors (IQ15), summarized by frequency:

- **Adding Prominent Reminders** (8 in total): The most common suggestion was visible notifications for privacy-related events. P9 suggested confirmation prompts before accessing sensitive data (e.g., clipboard), P16 advocated clearer, more user-friendly privacy interfaces, and P10 proposed placing personalized ads in a dedicated, optional section.
- **Reducing Personalized Advertising and Data Collection** (5 in total): Some suggested lowering ad frequency and data collection, as P6 noted: *"The personalized advertising provided is often irrelevant; I believe the app should offer more generalized content."*
- **Reducing Mandatory Permissions** (5 in total): Participants called for fewer forced permissions, as P1 noted: *"Many app compel me to consent to the upload and use of my private information; otherwise, access is denied."*
- **Incognito Mode / Data Deletion** (4 in total): Suggestions included privacy modes without tracking or periodic data deletion, as P21 proposed a *privacy forgetting* function to delete data after a short period.
- **Cross-App Privacy Restrictions** (3 in total): Some requested limits on cross-app data sharing, as P13 emphasized: *"If my phone number is leaked, there should be a way to trace which app collected it."*
- **Transparent Disclosure** (3 in total): Participants sought clarity on data use and recommendation logic, as P5 requested explanations of data, algorithms, and profiling.
- **Feature Enhancement** (3 in total): Some focused on improving recommendation quality, as P12 suggested: *"I hope the system not only recommends similar products repeatedly, but also suggestions aligned with my broader interests."*
- **Privacy Obfuscation** (2 in total): A few proposed reducing data precision, as P3 suggested generalizing location data to a coarse, kilometer-level range to enhance privacy.

Interpretation: Across these proposed mitigations, participants consistently asked for greater transparency and control over data collection. Prominent reminders, incognito modes, and clearer disclosures could help users understand when and why their data are collected, while reduced personalization, restricted cross-app data sharing, and fewer mandatory permissions could limit unwanted data collection. Proposals for

feature enhancement and data obfuscation further suggest that participants did not reject personalization itself, but preferred systems with clearer data practices and stronger user controls. Overall, these expectations reflect broader normative concerns about fairness, autonomy, and acceptable data practices. This aligns with prior work on contextual integrity [47], and contrasts with notice-and-choice models that assume disclosure or perceived benefits sufficiently legitimize data collection [48].

Finding I-5: Participants proposed measures emphasizing greater transparency, reduced reliance on personalized advertising and cross-app data sharing, and stronger user control over permissions and privacy settings.

VI. Quantitative Survey Methodology

Building upon preliminary interview findings, we developed and administered an online survey with 203 participants. Statistical analysis of the survey data enabled more robust conclusions regarding RQ1–RQ4 and validation of initial qualitative insights obtained from the interviews.

A. Research Model

Our survey design is grounded in the DEAL model [36]. The model proposes that information- and experience-driven factors contribute to *Surveillance Belief*, which is associated with *Perceived Surveillance* and subsequent *Surveillance* and *Advertising Responses* (§II-A3). We conceptualize users' beliefs in app e-eavesdropping as a concrete form of *Surveillance Belief*. Building on our interview findings, we further identify three factors not explicitly captured in the original model—*Objective Technical Literacy* (§V-B), *Trust in App Vendors* (§V-C), and *Privacy Cynicism* (§V-C, §V-D)—and extend the DEAL model accordingly (Figure 1).

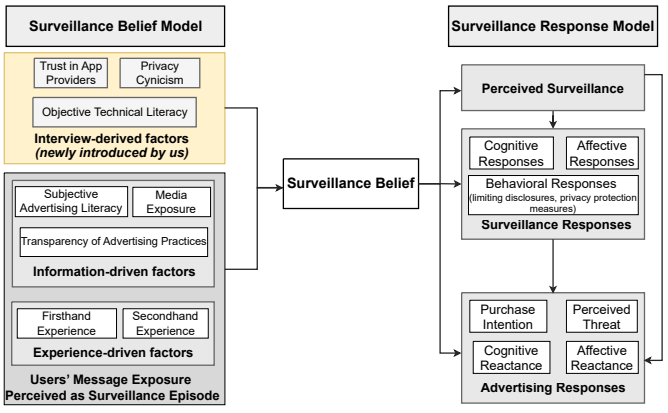


Fig. 1: Conceptual Framework of E-Eavesdropping Beliefs Formation and User Responses, adapted from the DEAL model [36]. Interview-derived factors are labeled as newly introduced by us; surveillance episodes refer to personalized content perceived by users as evidence of surveillance.

1) Surveillance Belief Model: The left side of Figure 1 illustrates the formation of *Surveillance Belief*. Following the DEAL framework, two types of factors contribute to surveillance beliefs: *information-driven* factors (e.g., transparency, media exposure, subjective advertising literacy) and *experience-driven* factors (e.g., firsthand and secondhand experiences). We integrated three *interview-derived extended factors* identified above, completing the set of influences in the Surveillance Belief Model:

(a) Dependent Variables. We employed *Surveillance Belief* as the dependent variable, representing users' cognitive belief regarding the existence of e-eavesdropping by apps.

(b) Independent Variables. Including the following factors: (❶) **Trust in App Vendors**: users' trust in vendors' responsible data handling [33, 86]; (❷) **Objective Technical Literacy**: users' knowledge of e-eavesdropping and personalized advertising [87, 88], proxied by understanding of personalization mechanisms, which played a more prominent role in belief change during the interviews (§V-B2). (❸) **Privacy Cynicism**: the belief that privacy protection is ineffective [83, 84]; (❹) **Transparency of Advertising Practices**: the extent of data disclosure, which can trigger surveillance beliefs ([89, 90]; (❺) **Media Exposure**: influence of privacy-related news coverage [91]; (❻) **Subjective Advertising Literacy**: general awareness of data use for advertising [92, 93]; (❼) **Firsthand Experience**: self-reported encounters with suspected e-eavesdropping [94, 95]; (❽) **Secondhand Experience**: awareness of such suspected incidents through others [95].

2) Surveillance Response Model: The right portion of Figure 1 illustrates the response pathway triggered by surveillance belief. Response variables include three components:

- **Perceived Surveillance:** refers to consumers' feeling of being monitored [33]. In the DEAL model, perceived surveillance is triggered by activated surveillance beliefs and by experiences interpreted as surveillance episodes.
- **Surveillance Responses:** responses to surveillance, including (1) *Cognitive Responses:* evaluating risks and benefits [82, 96]; (2) *Affective Responses:* emotional reactions to being monitored [97, 98]; and (3) *Behavioral Responses:* actions to limit exposure, such as *Limiting Disclosures* (reducing shared information [99]) and *Privacy Protection* (e.g., providing inaccurate data [92, 100]).
- **Advertising Responses:** reactions to advertising, including (1) *Purchase Intention:* likelihood of purchase [98, 101]; (2) *Perceived Threat:* feelings of manipulation [101]; (3) *Cognitive Reactance:* resistance to ad content [87]; and (4) *Affective Reactance:* emotional aversion [101].

While the DEAL model acknowledges various individual factors influencing these responses, these factors are not central to our research focus. In this study, we therefore focus on examining only the outcomes of users' responses to surveillance beliefs about app e-eavesdropping.

B. Survey Development and Organization

1) Survey Development: We developed the survey based on a comprehensive review of prior work on each factor, particu-

larly studies grounded in the DEAL model, adapting validated measurement scales to the context of e-eavesdropping and personalized advertising (Appendix G). A pilot study with 30 participants refined the instrument by clarifying ambiguous items and improving explanations, resulting in an average completion time of 15 minutes; accordingly, two attention-check items and a minimum completion time of 7 minutes were introduced to ensure response quality. An informed consent procedure emphasized voluntary participation, withdrawal rights, and data confidentiality for academic use.

2) Survey Organization: The final survey (Appendix H) consisted of three parts. The first collected demographic information (e.g., gender, age, education, occupation, CS background, income, and mobile OS). After introducing a brief definition of e-eavesdropping, the second measured first-hand experience, secondhand experience, and media exposure. The third assessed perceptions and beliefs regarding e-eavesdropping and personalized advertising using 7-point Likert scales (higher scores indicate stronger endorsement) covering all constructs in §VI-A. It also measured objective technical literacy through a multiple-choice item focusing on personalization mechanisms, which played a more prominent role in belief changes during interviews (§V-B2). All items were adapted from validated scales (Tables X and XI in Appendix G) to ensure construct validity and comparability. Although item wording followed target constructs, risk-oriented phrasing may have increased privacy salience and potentially primed responses, which is discussed in §VIII-C.

C. Participant Recruitment

Participants were recruited via *Credamo* [102], a professional survey platform in China, using its pre-screened panel to ensure representativeness of smartphone users aged 18+. No significant differences were found in Chi-square tests from the general smartphone user population in gender ($\chi^2(1) = 0.21$, $p = 0.645$) or age ($\chi^2(4) = 7.8$, $p = 0.112$) [37]. However, the sample was relatively well-educated and included a substantial proportion of participants with a CS background, which may limit generalizability. Eligibility required participants to be smartphone users aged 18+. Quality control included pre-screening, attention checks, and removal of speeders (<7 minutes). Of 250 responses, 203 were valid, with 25.6% aged 28–37 and 48.8% female. Participants received about 5 USD compensation. The study was IRB-approved, and full demographics are reported in Appendix I.

D. Data Analytical Approach

We employed two analytical approaches. For the survey's second part and objective technical literacy items in the third part, descriptive statistics summarized overall perceptions. For the *Surveillance Response Model*, descriptive analysis examined the strength of user attitudes and responses to perceived e-eavesdropping. For the *Surveillance Belief Model*, we conducted ordinary least squares (OLS) multiple linear regression to identify influencing factors. Reliability was assessed using Cronbach's α (threshold = 0.7) [103, 104]. Multicollinearity was tested via VIF, excluding variables with VIF > 5 [105]. Standardized coefficients were used to compare predictor effects, with significance set at $p < 0.05$.

VII. Quantitative Survey Findings

In this section, we present corresponding questionnaire results using both descriptive and inferential statistical analyses. Survey questions are denoted with the prefix "SQ" corresponding to items in Appendix H.

A. Descriptive Statistics: Users' Exposure, Belief of E-Eavesdropping and Objective Technical Literacy (RQ1, RQ2)

1) Self-Reported E-Eavesdropping Experiences and Surveillance Beliefs (SQ8-SQ11): We first assessed participants' exposure to perceived e-eavesdropping. Among 203 valid responses, 157 (77.3%) reported *firsthand* experiences, 157 (77.3%) *secondhand* experiences, and 181 (89.2%) *media* exposure. Table II shows that most exposures occurred *within the past month* (45.9% firsthand, 45.2% secondhand, 42.0% media), while notably, 27.4% of firsthand experiences occurred *within the past week*. Detailed statistics are provided in Appendix J. Surveillance belief was operationalized using three Likert-scale items: "App e-eavesdropping is real," "App e-eavesdropping is technically feasible," and "Certain apps are e-eavesdropping." Results in Table I show average scores between 5–6, with 74.4% of responses of surveillance belief in the 5–7 range.

Interpretation: Consistent with §III, participants reported both frequent recent experiences perceived as e-eavesdropping and strong surveillance beliefs. This quantitatively corroborates Finding I-1, which identified firsthand experiences and socially circulated accounts as key sources of such beliefs. The high prevalence and recency of these self-reported experiences may normalize surveillance interpretations and further entrench e-eavesdropping beliefs. Taken together, the qualitative findings establish experiential salience and social reinforcement as mechanisms of belief formation, while the survey confirms that both perceived experiences and strong beliefs are widespread within the sample.

Finding II-1: A large proportion of participants reported recent experiences of suspected e-eavesdropping, and most believe it exists and is actively conducted by apps; both findings corroborate Finding I-1.

2) Users' Objective Technical Literacy (SQ15): Based on interview findings in §V-B2, we operationalized objective technical literacy using eight true/false items adapted from prior measures [87, 88, 106], focusing on personalized advertising mechanisms (§II-C). The items cover personalization mechanisms identified in interviews, including cross-app tracking (Items 1, 5), cross-device linkage (Item 3), background interaction logging (Item 2), input-method data collection (Item 4), social-proximity propagation through close contacts (Item 6), and local-network or geospatial association mechanisms (Items 7, 8), which are often misattributed as e-eavesdropping. Content validity was ensured through review

TABLE I: Distribution of App E-Eavesdropping Surveillance Beliefs ($n = 203$, 1 = Strongly Disagree; 7 = Strongly Agree)

Statement	1	2	3	4	5	6	7	Mean	Median
App e-eavesdropping is real	2 (1.0%)	15 (7.4%)	10 (4.9%)	4 (2.0%)	59 (29.1%)	74 (36.5%)	39 (19.2%)	5.37	6.00
App e-eavesdropping is technically feasible	9 (4.4%)	8 (3.9%)	14 (6.9%)	18 (8.9%)	62 (30.5%)	49 (24.1%)	43 (21.2%)	5.14	5.00
Certain apps are e-eavesdropping	5 (2.5%)	17 (8.4%)	10 (4.9%)	16 (7.9%)	50 (24.6%)	61 (30.0%)	44 (21.7%)	5.21	6.00
Statement	**1-2**	**2-3**	**3-4**	**4-5**	**5-6**	**6-7**	**7**	**Mean**	**Median**
Overall Surveillance Belief	6 (2.9%)	15 (7.4%)	7 (3.4%)	24 (11.8%)	77 (37.9%)	65 (32.0%)	9 (4.4%)	5.24	5.67

TABLE II: Timing of Participants' Most Recent App E-Eavesdropping Exposure

Time Frame	Firsthand Experience (N = 157)	Secondhand Experience (N = 157)	Media Exposure Experience (N = 181)
Within the past week	43 (27.4%)	30 (19.1%)	21 (11.6%)
Within the past month	72 (45.9%)	71 (45.2%)	76 (42.0%)
Within the past 3 months	25 (15.9%)	37 (23.6%)	52 (28.7%)
Within the past 6 months	14 (8.9%)	13 (8.3%)	20 (11.0%)
1–2 years ago	2 (1.3%)	1 (0.6%)	2 (1.1%)
More than 2 years ago	–	–	–
I can't remember	1 (0.6%)	5 (3.2%)	10 (5.5%)

by two researchers specializing in personalized systems prior to deployment. Correct answers were scored as 1 (incorrect = 0). Table III reports descriptive statistics: participants showed relatively strong understanding (mean = 6.28/8, SD = 2.19), with accuracy above 90% for common mechanisms (e.g., tracking, logging) but lower (62–72%) for indirect mechanisms (e.g., social propagation, cross-device linkage), indicating misconceptions about latent data associations.

Interpretation: These results quantitatively corroborate the technical-awareness patterns observed in §V-B2: participants demonstrated substantial knowledge of personalization mechanisms, though understanding of indirect data associations remained weaker. Importantly, greater knowledge of these mechanisms did not consistently weaken e-eavesdropping beliefs. As observed qualitatively, awareness of extensive tracking, linkage, and inference capabilities may broaden users' perceptions of platform surveillance capacity and reinforce existing suspicions. This suggests that technical knowledge can increase perceived system capability rather than simply supply alternative explanations for ambiguous personalization—a pattern that motivates our subsequent analysis of the relationship between objective technical literacy and surveillance beliefs.

Finding II-2: Users show strong knowledge of common privacy risks but weaker understanding of indirect data propagation, indicating persistent misconceptions about latent data associations.

B. Inferential Statistics: Predictors of Surveillance Belief in App E-Eavesdropping (RQ3)

We constructed regression models for the independent and dependent variables in the surveillance belief model (SQ11-SQ17). The specific constructs and their corresponding items are presented in Table X in Appendix G.

TABLE III: Objective Knowledge Assessment Items and Descriptive Statistics ($n = 203$)

Statements (Correct Answer in Brackets)	Mean	SD	Correct n	Correct Rate (%)
The advertisement content I receive in one app may be personalized based on my usage behavior in other apps or on different mobile devices. [T]	0.91	0.29	185	91.1
App vendors are aware of the content I browse within their apps because many of my interactions are tracked in the background. [T]	0.90	0.30	183	90.1
If I have viewed a brand on one phone, it is possible that the company will display related advertisements on another phone I use. [T]	0.72	0.45	147	72.4
The words I type using a mobile input method may be collected and used to deliver targeted advertisements. [T]	0.90	0.30	183	90.1
When I see an advertisement for a brand in one app, it is likely due to my recent activity related to that brand in another app. [T]	0.91	0.29	184	90.6
Receiving advertisements similar to what my family or friends searched for is just a coincidence. [F]	0.62	0.49	125	61.6
Apps cannot detect other devices' behavior on the same Wi-Fi network and thus cannot recommend advertisements based on it. [F]	0.69	0.46	140	69.0
It is impossible for companies to show related ads on other devices in the same geographic area based on my own searches. [F]	0.63	0.48	128	63.1
Total Objective Literacy Score (0–8)	**6.28**	**2.19**	–	–

1) Reliability and Validity Checks: Reliability results are shown in Table XVIII[1] in Appendix K. Cronbach's α values (0.811–0.949) exceeded 0.7, indicating good internal consistency. Tables XVII in Appendix K report multicollinearity tests for variables related to *surveillance belief*: all VIFs were below 5 and correlations below 0.8, indicating no multicollinearity concerns.

2) Regression Analysis Results: Table IV presents OLS regression results for *surveillance belief*. The model shows good fit ($R^2 = 0.673$; adjusted $R^2 = 0.651$), explaining about 70% of the variance. Significant predictors include *subjective advertising literacy*, *objective technical literacy*, *privacy cynicism*, *firsthand experience*, and *secondhand experience*. Within the DEAL framework, both *firsthand* ($\beta = 0.191$, $p = 0.004$) and *secondhand experience* ($\beta = 0.214$, $p = 0.001$) positively predict surveillance belief, while *subjective advertising literacy* is the strongest factor ($\beta = 0.226$, $p < 0.001$). In contrast, *media exposure* and *transparency of advertising practices* are

Firsthand Experience and *Secondhand Experience* were single-item measures, and *objective technical literacy* was a cumulative score; thus, reliability testing was not applicable.

not significant. Among our extended factors extracted from the interview, *objective technical literacy* ($\beta = 0.212$, $p < 0.001$) and *privacy cynicism* ($\beta = 0.219$, $p < 0.001$) positively influence surveillance belief, whereas *trust in app vendors* has a negative effect ($\beta = -0.174$, $p = 0.023$).

TABLE IV: Multiple Regression Result

Construct	Standardized (β)	P$>\|t\|$
Firsthand Experience	0.191	0.004
Secondhand Experience	0.214	0.001
Media Exposure	0.056	0.351
Transparency of Advertising Practices	0.037	0.603
Subjective advertising literacy	0.226	0.000
objective technical literacy	0.212	0.000
Privacy Cynicism	0.219	0.000
Trust in App Vendors	-0.174	0.023
R^2	0.673	
Adjusted R^2	0.651	

Interpretation: These results identify multiple factors associated with e-eavesdropping beliefs. Firsthand and secondhand experiences were positively associated, whereas media exposure and perceived transparency were not significant predictors, highlighting the primacy of personally grounded encounters. Consistent with attribution theory, such experiences provide salient evidence for inferring e-eavesdropping when underlying mechanisms are opaque [81]. Both subjective and objective technical literacy also showed positive associations—a pattern contrasting with prior work suggesting literacy reduces misconceptions [40, 41]. Instead, awareness of extensive data collection and inference capabilities may expand perceived surveillance capacity and reinforce existing beliefs, aligning with research on folk theories and algorithmic imaginaries [29, 43–45]. Privacy cynicism was positively associated, while trust in app vendors was negatively associated, consistent with privacy cynicism as resignation to data misuse [83–85]. This suggests e-eavesdropping beliefs reflect not only technical understanding and experience, but also broader expectations about how platforms manage user data.

Finding II-3: Among the extended factors (Findings I-2 and I-3), *objective technical literacy* and *privacy cynicism* positively influence e-eavesdropping beliefs, while *trust in app vendors* has a notable negative effect.

C. User Responses to E-Eavesdropping Beliefs (RQ4)

Descriptive analysis of user responses to surveillance beliefs (SQ18–SQ26) is summarized in Table V, with constructs detailed in Appendix G and reliability reported in Appendix L. Participants were grouped into *low* ($\leq$3), *medium* (3–5), and *high* ($\geq$5) belief levels. These groups were compared across eight response variables introduced in §VI-A2. Across these groups, *perceived surveillance* increased markedly (mean: 1.97 to 5.66). This confirms that stronger e-eavesdropping beliefs are associated with greater perceived surveillance and clear polarization between skeptical and convinced users.

In surveillance responses, *cognitive responses* remained stable (3.7–4.0), whereas *affective responses*, *limiting disclosures*, and *privacy protection* increased with belief. *Limiting disclosures* levels were consistently lower than *privacy protection*, suggesting users prefer active safeguards over reduced expression. In the advertising response dimension, *purchase intention* was highest among low-belief users (mean = 5.02, median = 5.50) and declined among high-belief users (mean = 4.57, median = 5.33), indicating reduced advertising acceptance with stronger beliefs. In contrast, *perceived threat*, *cognitive reactance*, and *affective reactance* all increased with belief (e.g., Perceived threat: 1.80 low, 3.71 high; Cognitive reactance: 2.26 low, 3.73 high; Affective reactance: 1.76 low, 3.52 high), suggesting stronger defensive and emotional resistance.

Interpretation: These patterns indicate that stronger surveillance beliefs are associated with amplified emotional, behavioral, and advertising-related responses. While cognitive evaluations remained relatively stable, affective responses increased more substantially, suggesting that stronger beliefs about covert listening are tied more to feelings of intrusion than to rational reassessment—consistent with findings that opaque data practices elicit negative emotions [29]. This aligns with the DEAL framework, in which surveillance beliefs link to affective and behavioral surveillance responses, as well as downstream advertising responses [36]. Stronger beliefs also corresponded with greater privacy-protective actions and disclosure restriction. In the advertising domain, heightened perceived threat and reactance, alongside reduced purchase intention, point to diminished persuasive effectiveness.

TABLE V: Comparison of Mean and Median Scores across Different Levels of Surveillance Belief (↑ indicates positive responses, ↓ indicates negative responses)

Constructs	Low Belief ($\leq$3, N = 22)		Medium Belief (3–5, N = 44)		High Belief ($\geq$5, N = 151)	
	Mean	Median	Mean	Median	Mean	Median
Perceived Surveillance (↓)	1.97	1.87	4.52	5.00	5.66	5.75
Surveillance Responses – Cognitive Responses (↑)	3.86	3.00	3.98	3.67	3.71	3.33
Surveillance Responses – Affective Responses (↓)	1.76	1.67	3.02	2.67	3.74	3.00
Surveillance Responses – Limiting Disclosures (↓)	2.32	2.25	4.03	3.75	4.69	5.25
Surveillance Responses – Privacy Protection (↓)	3.94	3.40	4.52	5.20	5.13	5.60
Advertising Responses – Purchase Intention (↑)	5.02	5.50	4.79	5.33	4.57	5.33
Advertising Responses – Perceived Threat (↓)	1.80	1.63	3.37	2.75	3.71	3.25
Advertising Responses – Cognitive Reactance (↓)	2.26	2.00	3.62	3.67	3.73	3.67
Advertising Responses – Affective Reactance (↓)	1.76	1.75	3.07	2.50	3.52	2.75

Finding II-4: Stronger e-eavesdropping beliefs are associated with greater perceived surveillance, negative affect, perceived threat, and reactance, as well as increased privacy protection and disclosure limitation, and lower purchase intention—indicating broad emotional and behavioral consequences, consistent with Finding I-4.

VIII. Discussion

A. Summary of Findings and Comparison with Prior Work

Our mixed-method findings offer a coherent account of e-eavesdropping belief formation and its consequences. Interviews revealed that participants frequently grounded their beliefs in firsthand experiences, social anecdotes, and uncertainty about app data practices. These beliefs were accompanied by negative emotions and limited privacy-protective actions. These insights informed our extension of the DEAL framework [36] with three constructs: distrust, privacy cynicism, and objective technical literacy. Survey results quantitatively supported these patterns: firsthand and secondhand experiences were widespread (77.3% each); participants understood common tracking but were less knowledgeable about indirect personalization; firsthand and secondhand experiences, objective technical literacy, and privacy cynicism were positively associated with e-eavesdropping beliefs, whereas trust in app vendors was negatively associated. Notably, interviews showed that alternative-personalization knowledge more often reinforced beliefs, while e-eavesdropping knowledge had mixed effects—a pattern mirrored in the survey, where the personalization-related component of technical literacy was positively associated with belief. Stronger beliefs also correlated with greater perceived surveillance, negative affect, privacy-protective responses, threat, reactance, and lower purchase intention.

These findings both corroborate and extend prior work (Table VI in Appendix A). Consistent with Segijn et al. [1], we observe prevalent firsthand and secondhand experiences and stronger negative affect; Our mixed-method design further examines their roles in belief formation and downstream responses in the underexplored Chinese context. While Segijn et al. [34] did not examine technical literacy as a factor shaping surveillance beliefs, we objectively measure technical literacy and show that knowledge of alternative personalization mechanisms is positively associated with e-eavesdropping beliefs, often reinforcing rather than reducing them. Frick et al. [33] quantitatively identify psychological and trust-related predictors of perceived conversation surveillance; we extend this work by introducing interview-derived privacy cynicism and objective technical literacy, and show that personalization knowledge can paradoxically reinforce e-eavesdropping beliefs. Finally, building on Dunbar et al. [35], we combine qualitative interviews with a quantitative survey to examine how beliefs about unverified app e-eavesdropping form.

B. User-Proposed Mitigation Approaches

Participants proposed diverse mitigations, including prominent notifications, reduced personalization, fewer permissions, incognito modes, cross-app restrictions, transparent disclosures, and data obfuscation. These proposals primarily reflect participants' expectations about how privacy controls should work rather than their technical feasibility. Some aim to restore perceived control (e.g., dedicated ad sections), consistent with prior work [40, 41], while others (e.g., data tracing) express deeper distrust and demand institutional accountability. From a security perspective, feasibility varies:

- **UI-Level Controls:** Notifications and clearer permissions are highly feasible and already supported but mainly reduce perceived creepiness rather than actual tracking.
- **Data Minimization:** Reducing personalization and data collection is technically possible but constrained by platform incentives, as many services rely on precise targeting.
- **Retention-Limiting Modes:** Incognito and deletion are feasible locally but difficult to enforce across distributed systems and third-party data brokers without auditing.
- **Cross-App Restrictions and Tracing:** Restricting or tracing cross-app data flows would require broader platform-level support and coordination among multiple parties.

Overall, these mitigations may increase visibility and control, but they reduce ambiguity and mistrust rather than confirm or refute e-eavesdropping. They also reveal a mismatch between user expectations and the complexity of inference-driven advertising. Addressing this mismatch requires clearer information about data collection and personalization. Designers could make data practices more visible and verifiable; industry and regulators could strengthen accountability for cross-app flows and data provenance; researchers could audit acoustic and non-acoustic data pathways and develop literacy materials that distinguish technical feasibility, inference-based personalization, and verified deployment.

C. Limitations

1) Cross-Cultural Context: Our data were collected primarily in China, where cultural and institutional privacy contexts may differ from other regions; findings should not be assumed to generalize across cultural contexts.

2) Participant Generalizability: Campus-based recruitment overrepresents highly educated participants. However, our goal was theory-driven investigation of belief formation, not prevalence estimation; a tech-aware sample facilitates examining technical literacy and reasoning. Similar perceptions appear in broader populations: Frick et al. [33] included 58.8% university graduates and most having at least a high-school education, while Segijn et al. [1, 34] studied broader populations with only 7.2–11.7% students. Both reported substantial eavesdropping perceptions, suggesting the phenomenon is not limited to highly educated users. Nevertheless, our prevalence estimates should not be generalized to the broader population.

3) Xiaohongshu Sampling and Model Bias: Keyword-based collection may overrepresent users who explicitly discuss e-eavesdropping, and GPT-assisted classification may be sensitive to prompt wording or salient themes. We mitigated these risks through repeated extraction, manual refinement, and validation against 300 manually annotated posts ($k = 0.81$). The resulting taxonomy characterizes the collected posts, not the broader Xiaohongshu population.

4) Interview and Survey Biases: (a) *Self-Reported Experience Bias.* Self-reported experiences reflect perceptions, not verified events, and may be subject to recall bias, subjective interpretation, and misattribution. The binary interview measure

lacked detailed contextual probes (e.g., ad timing, concurrent activities), limiting our ability to distinguish perceived e-eavesdropping from coincidence or behavior-driven targeting. Future work should collect more event-level contextual data. (b) *Briefing and Survey Framing Bias.* The technical briefing may increase the salience or perceived plausibility of unfamiliar attacks despite our distinction between feasibility and deployment. The survey may similarly be subject to priming effects, as risk-oriented wording could heighten privacy concerns. Future studies could employ neutral wording and assess briefing-related effects. (c) *Cognitive Load.* No apparent cognitive overload was observed during interviews, but load was not directly measured and cannot be fully excluded.

IX. Conclusion

This study examines e-eavesdropping beliefs and their consequences among Chinese users. Beliefs were associated with perceived experiences, privacy cynicism, distrust in app vendors, and technical literacy, with greater knowledge of personalization mechanisms sometimes coinciding with stronger beliefs. Stronger beliefs were associated with negative affect, privacy-protective responses, and lower purchase intention, although protective action remained limited. Participants called for greater control and transparency. Overall, technical knowledge alone may not reduce e-eavesdropping beliefs; clearer explanations of data collection and inference, together with stronger user controls, may help users distinguish inference-based personalization from unverified e-eavesdropping.

Ethics Considerations

This study, involving surveys and interviews on perceived app e-eavesdropping, was approved by our university's Institutional Review Board and followed established ethical guidelines [107]. All participants voluntarily provided informed consent, were informed of data usage, and could skip questions or withdraw at any time. We minimized data collection, pseudonymized responses, securely stored research data with restricted access, and reported only aggregated results and de-identified quotations without releasing sensitive raw data. Potential discomfort from discussing privacy concerns was mitigated through voluntary participation and avoidance of highly sensitive questions. To prevent misinterpretation, we distinguish perceived e-eavesdropping from verified surveillance and avoid unsupported claims. While our findings may inform user awareness, developer transparency, and policymaking, we acknowledge the risk of heightened privacy concerns and present results responsibly, emphasizing user protection and balanced interpretation.

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APPENDIX

All appendices, including additional methodological details, supplementary analyses, and supporting results, are provided in the supplementary material [108].